

# Multi-Agent System: Theories and Applications

## Lecture 14: A Glimpse on More Advanced Topics

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# Outline

- 1 Time Delays
- 2 Heterogeneity
- 3 Nonlinearity
- 4 System Constraints
- 5 Disturbances and Noises
- 6 Generalized Edge Dynamics

## Where delays come from?

### Input delay

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t-\tau)$$

$$y_i(t) = Cx_i(t)$$

### State delay

$$\dot{x}_i(t) = Ax_i(t-\tau) + Bu_i(t)$$

$$y_i(t) = Cx_i(t)$$

- self-delay on agents
- delay (latency) on inter-agent communication channels

# Sources of heterogeneity

## Heterogeneous linear MAS

$$\dot{x}_i(t) = A_i x_i(t) + B_i u_i(t - \tau_i)$$

$$y_i(t) = C_i x_i(t)$$

$$A_i \in \mathbb{R}^{n_i \times n_i}, B_i \in \mathbb{R}^{n_i \times m_i}, C_i \in \mathbb{R}^{p_i \times n_i}$$

- Models of agents are different, e.g., distinct dynamics, uncertainties, etc.
- Inter-agent communication channels have different characteristics, e.g., bandwidth, latency, etc.

# Sources of nonlinearity

## Example of nonlinearity caused by input saturation

$$\dot{x}_i(t) = Ax_i(t) + B\text{sat}(u_i(t))$$

$$y_i(t) = Cx_i(t)$$

- Agents' dynamics.
- Inter-agent communication channels.
- System constraints.

# What can be constraints?

- Physical limitations of agents, e.g, that of on-board sensors and actuators, limited capacity of batteries, etc.:  $\|x(t)\| \leq \alpha, \|u(t)\| \leq \beta$
- Limitations on inter-agent sensing and communication, e.g., channel bandwidth, speed, latency, etc.:  $\|x_i(t) - x_j(t)\| \leq \gamma$
- Network constraints, e.g., limited amount of memory, sampling rate, etc.

# What are effects of constraints?

- Limited freedom on system control design.
- Time-varying, random communication structure between agents: **time-varying  $\mathcal{G}(t)$** .
- Requirement for fast convergence and energy-effective control algorithms.

# Where they come from?

## Example of systems with disturbances and noises

$$\begin{aligned}\dot{x}_i(t) &= Ax_i(t) + Bu_i(t, \sigma_i(t)) + Ed_i(t) \\ y_i(t) &= Cx_i(t) + G\eta_i(t)\end{aligned}$$

- Surrounding environment.
- Physical attacks, sensor and actuator faults.
- Noises on communication channels.
- Cyber-attacks.



## What are generalizations of edge dynamics introduced in Lecture 8?

- Weighted graph.
- Directed graph.
- Nonlinear dynamics of agents.
- Nonlinear inter-agent coupling.